

Course: ITAI-1371

Professor: Viswanatha Rao

Brian Oommen, Nguyen Khanh Toan Chu, Montserrat Tarifa Capote

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This reflection discusses what our group learned from the Module 2 lab and how it improved our understanding of pandas, NumPy, and Matplotlib. The lab used the Iris dataset to demonstrate how Python can be used to organize, analyze, and visualize data. By completing the activities and assessment tasks together, we developed a better understanding of how these tools work and how they can be applied to machine learning.

The main focus of the lab was working with the Iris dataset using three Python libraries. Pandas was used to organize and examine the dataset in a DataFrame, NumPy was used for numerical calculations, and Matplotlib was used to create graphs and visualize the data. We worked through the notebook using Jupyter/VS Code and ran different code cells to see the results. The assessment tasks included calculating summary statistics, such as the mean and standard deviation, and creating a bar chart showing the number of each Iris species.

One of the main things we learned was how these libraries can work together during the data analysis process. Before completing the lab, we had some previous knowledge of pandas, NumPy, and Matplotlib, but we did not fully understand how they could be used together. Working with the Iris dataset made their different purposes clearer. Pandas helped us organize the data, NumPy helped us perform calculations, and Matplotlib helped us understand the data

through visualizations. Seeing the results after running the code made the concepts easier to understand.

Some parts of the lab were confusing at first, especially when we were learning how different functions worked. However, working together as a group helped us overcome these difficulties. We discussed the code, compared our understanding, and checked the output after each cell. This helped us understand that it is important not only to run code but also to examine and interpret the results. This was one of the most useful skills we developed during the lab.

The lab also helped us understand the importance of data analysis in machine learning. Before building a machine learning model, it is important to understand the dataset and identify patterns or differences in the data. Pandas can help organize and prepare datasets, NumPy can be used for numerical calculations, and Matplotlib can help visualize information. These tools therefore provide an important foundation for future machine learning activities.

We expect to use the skills from this lab in future labs and projects. As datasets become larger and more complex, being able to organize, analyze, and visualize data will become increasingly important. We also need to continue improving our ability to read code and understand its output rather than simply checking whether the code runs successfully.

Overall, the Module 2 lab gave our group practical experience with pandas, NumPy, and Matplotlib. It improved our understanding of DataFrames, statistical calculations, and data visualization while also showing us the importance of preparing and understanding data before applying machine learning techniques. Although some parts required practice, working together and reviewing the results helped us become more confident and prepared for future labs.